

Gaurav Gupta

SCIENTIST/ENGINEER-SC

Space Application Centre, Ahmedabad

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Summary

Aerospace engineer at the Space Applications Centre (ISRO) specializing in computational fluid dynamics and biological fluid mechanics. Primary experience encompasses investigating avian flight aerodynamics, modeling microgravity two-phase flows, and developing custom flow-solver subroutines in Fortran and Python, alongside engineering precision optomechanical systems for opto-quantum communication payloads. Key research interests focus on fluid mechanics of animal locomotion, biological flows, fluid-structure interactions, and low-gravity fluid mechanics.

Engineering Research Experience

Space Application Centre, Indian Space Research Organization (ISRO)

Ahmedabad, Gujarat

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Jul. 2025 - Present

Opto-Mechanical Engineering for Opto-Quantum Payloads

- Responsible for engineering space-qualified optomechanical subsystems for quantum communication payloads.
- Led precision assembly, integration, and sub-micron alignment protocols for sensitive quantum optics setups.
- Developed and implemented automated workflows and CAD/CAE pipelines for opto-mechanical design and STOP analysis using Siemens NX Open APIs.

Microgravity Fluid Mechanics

- Formulated the conceptual system design and architecture for a space-based recirculating aquatic life support payload.
- Conducted two-phase simulation to investigate the surface tension driven fluid flow and transport in microgravity environment using OpenFOAM.
- Investigating stability of an air-water interface under microgravity using OpenFOAM.

Indian Institute of Technology, Kharagpur

Kharagpur, India

RESEARCH INTERN

Mar. 2023 - May. 2025

- Explored various high-fidelity computational methods to investigate the aerodynamics of avian tails in gliding flight.
- High-fidelity DNS (for 2D) and hybrid RANS-LES methods such as DES were identified as suitable computational methods for investigating the aerodynamics of avian flight.
- Conducted high-fidelity DNS (using Incompact3D and NEK5000), DES, and RANS (using SU2) of a representative common swift geometry at Re 10000 - 20000.
- Pursued under the guidance of Dr. Sandeep Saha, IIT KGP and Dr. Manu K V, IIST.

Education

Indian Institute of Space Science and Technology

Thiruvananthapuram, India

B.TECH. IN AEROSPACE ENGINEERING

Dec. 2021 - Jun. 2025

- Cumulative Grade Point Average (CGPA): 8.28/10
- Relevant coursework: Aerodynamics, Computational Fluid Dynamics, Aircraft Design, Boundary Layer Theory, Finite Element Method, Flight Dynamics & Control, Introduction to Astrobiology
- Relevant Labs: Aerodynamics Lab, Modelling and Simulation Lab, Flight Mechanics Lab, and Programming

Kendriya Vidyalaya No.1 AFS Kalaikunda

Kharagpur, India

HIGH SCHOOL (XI & XII)

Jul. 2019 - May. 2021

- Graduated as the topper of school with 97.4% in the CBSE AISSEE examination.

Open-Source Contributions

Incompact3D

CFD, Fortran

HIGH-ORDER FINITE DIFFERENCE FLOW SOLVER

GitHub

- Developed a fortran subroutine to calculate lift and drag forces for a 3D immersed body simulations using Incompact3D.
- Working on improving the performance of importing STL files for high-fidelity simulations using Incompact3D.

PyMech

CFD, Python, NEK5000

A PYTHON SOFTWARE SUITE FOR NEK5000 AND SIMSON

GitHub

- Improving the in-built open_dataset function to support unstructured datasets produced using NEK5000.
- Used for post-processing of high-fidelity simulation data using NEK5000.

TurbCourse

CFD, Python, NEK5000

A COURSE IN TURBULENCE SIMULATION

GitHub

- Improved the performance of POD and DMD modules for high-fidelity simulation data using NEK5000.
- Added capability for conducting frequency analysis of the POD modes.
- Uses PyMech for reading/writing of NEK5000 data.

FreeFEM-Euler

CFD, FreeFEM++, GMSH

FREEFEM BASED INCOMPRESSIBLE EULER SOLVER.

Jun. 2024

- An incompressible inviscid flow solver based on Artificial Compressibility Method was developed using FreeFEM++.
- The code is capable of solving 2D problems based on in-built FreeFEM++ and GMSH meshes.
- Open-source and available on GitHub.

Skills

Modelling and Computer Aided Design

DS Solidworks, DS Catia, Autodesk Fusion 360, Autodesk Autocad, Blender, Siemens NX

Flow Solvers and Mesh Generation

Incompact3D, NEK5000, SU2, NEKTAR++, OpenFOAM, GMSH, Coreform Cubit, Pointwise

Programming and Numerical Libraries

C++, Fortran, FreeFEM++, Julia, Python, MATLAB

HPC and Tools

Linux, Git, LaTeX, Visit, Paraview

Publications

Enabling automation in Opto-Quantum Communication Program through APIs

Internal Proceedings/Non-Public

GUPTA, GAURAV, SHARMA, ANKIT

2026

ISRO/DOS Seminar on Computer Technology, Cyber Security and Mission IT 2026

Certificates

Ongoing **Fundamentals of Fluid-Solid Interactions**, École Polytechnique and Coursera

Mar 2025 **Graduate Aptitude Test - Aerospace Engineering**, Indian Institute of Technology, Roorkee

All India Rank 60

Apr 2024 **High Performance Computing**, IIT Bombay and NPTEL

Dec 2022 **Machine Learning with Python (Honors)**, IBM and Coursera

Oct 2022 **Design of Fixed Wing Unmanned Aerial Vehicle (Topper)**, IIT Kanpur and NPTEL

Nov 2022 **Aircraft Design**, IIT Bombay and NPTEL

Extracurricular

- Directed IIST's technical festival (Conscientia) for 1000+ participants as Chief Coordinator, managing a INR 2.2 million budget and securing an INR 400,000 profit.
- Led team initiatives in designing fixed-wing and rotor aircraft as Student Coordinator, while orchestrating UAV design workshops and competitions.